Physical Chemistry Cume
April 1 2011

For binding of two species (for example, the interaction of a protein, E, and a substrate, S) that is proposed to interact according to a simple 2-state mechanism:

$$E + S \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} ES$$

(a) What is the relationship between the association equilibrium constant, K_a, the dissociation equilibrium constant, K_d and the elementary rate constants k_{on} and k_{off}?
(10 pts)

(b) Show that when the binding is relatively weak, [ES] at equilibrium is given by

$$[ES] = \frac{[E]_0[S]_0}{K_d + [S]_0}$$

Where the subscript 0 refers to the concentration at time = 0.
(30 pts)

(c) Show that when the binding is relatively strong, [ES] at equilibrium is given by:

$$[ES] = \frac{[E]_0 + [S]_0 + K_d - \sqrt{([E]_0 + [S]_0 + K_d)^2 - 4[E]_0[S]_0}}{2}$$

(30 pts)

(d) Let us now extend the mechanism above so that, after binding, the substrate reacts to form a product, P

$$E + S \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} ES \xrightarrow{k_{cat}} E + P$$

(i) Under the conditions described in (b), What is the initial rate of formation of P, v_0, and what is the maximum rate, V_{max}, in terms of the elementary rate constants and the initial concentrations of [E] and/or [S]?

(ii) Under the conditions described in (c), What is the initial rate of formation of P, v_0, and what is the maximum rate, V_{max}, in terms of the elementary rate constants and the initial concentrations of [E] and/or [S]?

(30 pts)